

Algebra II - Tutorial 4 - Date: 19/2/2026

1. Let G be a group and let $\text{Inn}(G) := \{\phi_g : G \rightarrow G \text{ where } \phi_g(x) = gxg^{-1}; g \in G, x \in G\}$ denote the set of all inner automorphisms of G . Show that $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$.
2. Suppose $G/Z(G)$ is cyclic. Show that G is abelian.
3. Let G be a group of order pn where p is prime and $p > n$. Let H be a subgroup of G . Then H is normal in G .
4. Let G be a nonabelian group of order p^3 .
 - (a) What are the possibilities for $Z(G)$, the center of G .
 - (b) Let $x \in G \setminus Z(G)$. What is the order of $\langle x \rangle$.
 - (c) What are the possibilities of the class equation of G ?
 - (d) What is the class equation of (i) D_4 (ii) Q_8 ?

Let G be a group of order 12 and K a Sylow-3 subgroup. Consider $G/K = \{g_1K = K, g_2K, g_3K, g_4K\}$ and let $\text{Sym}(G/K)$ denote the symmetric group on the set G/K . Let $\phi : G \rightarrow \text{Sym}(G/K)$ be the map given by $\phi(g)(g_iK) := gg_iK$.

- (a) Show that ϕ is a homomorphism
 - (b) Show that ϕ is injective if and only if K is not a normal subgroup of G .
 - (c) If ϕ is injective, show that $G \cong A_4$.
5. Let $p \neq q$ be prime numbers and let G be a group of order p^2q . Show that G has a normal sylow subgroup.